Chunxiao (Calvin) Lin

chunxiaol@vt.edu | chunxiao-lin-2023.github.io | Google Scholar

Research Summary

FPGA researcher building **Echo State Network (reservoir computing) accelerators with on-chip learning** for **Next-G wireless**, from DSP-level microarchitecture to **real-time PC–FPGA testbeds** on live distributed-MIMO links. 12 publications, including first-author IEEE TCAD, ICCAD, ICNC, ISQED and Frontiers.

Education

Virginia Tech, Ph.D. in Computer Engineering — GPA: 4.0/4.0 — *Sep 2021 – May 2027 (Expected)*

- Advisor: Prof. Yang (Cindy) Yi | BRICCS@VT (Brain-Inspired Computing, Communication, and Security Lab)

University of Southern California, Master of Engineering — GPA: 3.95/4.0 — *Sep 2019 – May 2021*

- Focus: Computer Architecture, VLSI Design, Digital System Design

Technical Skills

Hardware: Verilog/SystemVerilog RTL, DSP48 datapaths, systolic arrays, fixed-point design; Xilinx Virtex-7, Zynq-7000, Zynq UltraScale+, RFSoC (ZCU216)

Systems: 10G/1G Ethernet (MAC, PCS/PMA), UDP/IP (lwIP), AXI, PS–PL co-design; USRP SDRs, GNU Radio, Keysight PROPSIM

Tools: Vivado, Vitis, Quartus, Cadence, ns-3; Python, MATLAB, C

Research Experience

Graduate Research Assistant, BRICCS@VT, Virginia Tech — *Sep 2021 – Present*

Real-Time PC–FPGA Testbeds for Distributed MIMO Channel Prediction — *Feb 2026 – Jul 2026*

- Built two end-to-end testbeds that close a live 4×2 DMIMO loop (4-antenna gNB, two 2-antenna UEs, USRP N310/X310, Keysight PROPSIM F16): a 1 GbE PS–PL hybrid (lwIP UDP/IP on the ARM) and a 10 GbE pure-RTL design with the Ethernet MAC/PHY and one windowed-ESN predictor per UE in programmable logic.
- **≤0.9 ms end-to-end latency** and **1,250 predictions/s per UE** at 3.42 W; NMSE 0.0119 (outdated CSI) → 0.0087 in live operation.

FPGA-based ESN with On-chip RLS Learning for MIMO Channel Prediction — *IEEE TCAD 2026* — *Apr 2024 – Dec 2024*

- Architected the first complete on-chip ESN with recursive-least-squares training on FPGA: a DSP48 systolic-array readout and an inversion-free, three-stage pipelined RLS engine whose latency grows only linearly with the state size.
- **3.87 μs per training + inference step**, 264 Mb/s at 1.2 W: **>1600× / 290× the energy efficiency** of CPU / GPU, and 5.3× lower NMSE than the prior on-chip FPGA learner.

High-Speed Reservoir Neuron and HW/SW Co-design — *ICNC 2026* — *Jun 2024 – Sep 2025*

- Designed a hybrid DSP–CLB neuron (nine DSPs + carry-save adder tree) and a range-addressable tanh LUT: **+25% DSP efficiency**, 13% shorter processing time, 25% less tanh memory; the full ESN runs at **65.98 MS/s** and beats AR prediction by 5.7–14.2 dB in NMSE.
- Ported the ESN to a Zynq-7000 SoC with PL-based inference and PS-based learning.

ESN Symbol Detection for (Massive) MIMO-OFDM on FPGA — *ICCAD 2023, Frontiers 2024* — *Jun 2022 – Dec 2023*

- Designed a nine-DSP48 reservoir neuron and built an SDR/FPGA platform (USRP, GNU Radio, Virtex-7) for over-the-air 2×2 MIMO-OFDM detection: **3.3× the throughput** of the prior SISO accelerator, hardware BER within 0.03% of software and below MMSE in every scenario.
- Extended the detector to massive MIMO, outperforming LMMSE from 2×2 up to 4×64 antennas with only 8–16 reservoir neurons.

Optimized Reservoir Node Architecture for FPGA-based RC — *ISQED 2022* — *Oct 2021 – Feb 2022*

- Bit-serial, CSD-encoded neuron that hard-wires the fixed sparse weights: **80.7% fewer LUTs** and DSPs cut from 162 to 16 versus a DSP-based ESN.

Other Projects

Low-Power 10T-SRAM for In-Memory Computing (USC) *Oct 2020 – Dec 2020*

- 64×64 10T-SRAM array with two asymmetric sense amplifiers for in-memory NOR/NAND; leakage cut with high-Vt devices, sizing, stacking and negative Vgs.

Dynamically Secured Scan Chain and Scan-based Attack (USC) *DFT 2021* *Aug 2020 – Nov 2020*

- Python netlist-to-Verilog flow with dynamic encryption logic and ModelSim verification of the attack model; contributed to GF-Flush (500-bit keys in <7 s, $>4300\times$ faster than SAT attacks).

Publications

- **C. Lin**, U. S. Khan, M. Zheng, M. F. Azmine, L. Liu, D. J. Jakubisin, and Y. Yi, "FPGA-based Reservoir Computing with On-chip Learning for MIMO Channel Prediction," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2026.
- S. S. Chowdhury, M. R. Sarkar, J. Ling, **C. Lin**, and Y. Yi, "SPARC: Spiking Plasticity-Adaptive Reservoir Computing with On-Chip Reward-Modulated Triplet STDP for Spectrum Sensing," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2026, accepted.
- **C. Lin**, U. S. Khan, L. Liu, and Y. Yi, "FPGA-based High-speed Reservoir Computing for MIMO Channel Prediction," in *Proc. International Conference on Computing, Networking and Communications (ICNC)*, 2026.
- R. Gaurav, S. Sinha, **C. Lin**, T. C. Stewart, L. Liu, and Y. Yi, "Multivariate Legendre-SNN on Loihi-2 for Time Series Classification and 5G Jamming Detection," *IEEE Transactions on Circuits and Systems for Artificial Intelligence*, 2026.
- A. Liu, M. F. Azmine, **C. Lin**, and Y. Yi, "Reservoir Computing: Foundations, Advances, and Challenges Toward Neuromorphic Intelligence," *AI*, vol. 7, no. 2, 2026.
- A. Liu, **C. Lin**, D. Rosen, and Y. Yi, "Echo State Networks in Reservoir Computing: Foundations, Benchmarks, and Applications to Next-G Wireless Communication," in *Proc. IEEE International Conference on Data Mining Workshops (ICDMW)*, 2025.
- **C. Lin** and Y. Yi, "Enhancing FPGA CAD Flow with AI-Powered Solutions," in *AI-Enabled Electronic Circuit and System Design*, Springer, 2025.
- K. Yang, Y. Hu, D. Chen, **C. Lin**, Y. Yi, P. A. Beerel, and P. Nuzzo, "Machine Learning-Enhanced Analysis and Design for Trustworthy Integrated Circuits," in *AI-Enabled Electronic Circuit and System Design*, Springer, 2025.
- **C. Lin**, M. F. Azmine, Y. Liang, and Y. Yi, "Leveraging Neuro-Inspired AI Accelerator for High-Speed Computing in 6G Networks," *Frontiers in Computational Neuroscience*, vol. 18, 2024.
- **C. Lin**, M. F. Azmine, and Y. Yi, "Accelerating Next-G Wireless Communications with FPGA-Based AI Accelerators," in *Proc. IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2023.
- **C. Lin**, Y. Liang, and Y. Yi, "FPGA-based Reservoir Computing with Optimized Reservoir Node Architecture," in *Proc. International Symposium on Quality Electronic Design (ISQED)*, 2022.
- D. Chen, **C. Lin**, and P. A. Beerel, "GF-Flush: A GF(2) Algebraic Attack on Dynamically Secured Scan Chains," in *Proc. IEEE International Symposium on Defect and Fault Tolerance in VLSI and Nanotechnology Systems (DFT)*, 2021.